Review notes · for you

Spreadsheets and clean data: unit check

Data engineering · Level 2 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Cells, rows and formulas
- Messy names, wrong counts
- Blanks, repeats and impossible values
- Summaries that tell the truth

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (15 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (8 min): Reveal each answer and ask one team to explain why it's right.

Discussion

- Which cleaning step do you think matters most?
- Where could you use a median instead of a mean?
- What would you check before trusting a table?

Check yourself

- One thing I'll check before I trust a number, and why.

Try it this week

Take one table you made in this unit: check it for blanks, repeats and impossible values, then sum it up with the right summary and an honest chart.

Quiz sheet

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Name _____ Class _____ Date _____

1 **What does =AVERAGE(C2:C9) work out?**

◯ The mean of every cell from C2 to C9 ◯ The mean of C2 and C9 only ◯ The total of row 2

2 **Fill the gaps: print the total and how many marks there are.**

```
marks = [12, 15, 9]
print( _____ (marks), _____ (marks))
```

Word bank: `max` `sum` `sorted` `len`

3 **Which of these would a computer count as different from "football"? Choose all that apply.**

☐ "Football" ☐ "football" ☐ "football " (space at the end)

4 **Put the cleaning steps in order.**

☐ Match to the allowed list

☐ Count again

☐ Remove the spaces at the ends

☐ Make the letters match

5 **A blank mark should be counted as zero in the average.**

◯ True ◯ False

6 **This should print 1, the number of blanks, but it prints 2. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  marks = [14, None, 0, 9, 0]
2  blanks = 0
3  for m in marks:
4      if m == 0:
5          blanks = blanks + 1
6  print(blanks)
```

7 **Match each problem to what you should do.**

A mark of 30 out of 20	• • • •	Say it’s missing
The same reading sent twice	• • • •	Check it with the teacher
A blank reading	• • • •	Keep one copy

8 **Marks are 10, 11, 12, 13 and 99. Which summary describes them best?**

◯ The median: 12 ◯ The mean: 29 ◯ The range: 89

9 **Complete the sentence.**

A fair bar chart starts its bars at _____ and has a _____.

Word bank: fifty · zero · title · secret

10 **What does this program print?**

```
data = [8, 2, 9, 5, 7]
print(sorted(data)[2])
```

◯ 7 ◯ 5 ◯ 8

Check your answers

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1 **The mean of every cell from C2 to C9**

C2:C9 is a range: every cell from C2 to C9.

2

```
marks = [12, 15, 9]
print(sum(marks), len(marks))
```

sum adds the marks (36) and len counts them (3).

3 **"Football"; "football " (space at the end)**

Only an exact copy matches. A different letter or a space at the end makes a new answer, so tidy the text before counting.

4 **Top to bottom: 3, 4, 1, 2**

Tidy first, then match, then count: each step makes the next one work better.

5 **False**

A blank means nobody recorded a mark. Counting it as zero pulls the average down with a mark nobody got.

6 **Line 4: `if m is None:`**

0 is a real mark; blanks are None, so the test must be m is None.

7 **A mark of 30 out of 20 — Check it with the teacher**
The same reading sent twice — Keep one copy
A blank reading — Say it's missing

Impossible values are checked, copies are removed, and blanks are reported as missing, never guessed.

8 **The median: 12**

99 is far from the rest, maybe a typo. It drags the mean up to 29, but the median stays at 12, like most of the marks.

9 **A fair bar chart starts its bars at zero and has a title.**

Bars from zero show the real differences, and a title says what the chart is about.

10 **7**

sorted puts the list in order: 2, 5, 7, 8, 9.